

Randy Tang

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SUMMARY

Software engineer with 3+ years building production-critical data systems at a global bank (OTC trade reporting, tens of millions of daily transactions). Full-stack builder across Python pipelines, React/TypeScript dashboards, and LLM-powered applications. Currently co-building a deployed AI platform for a consulting agency — embedded with the founder, translating business needs into shipped tools at speed. Cornell CS & ME.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Java (fundamentals) | **Frameworks/Infra:** Next.js, React, Flask, FastAPI, Supabase, Postgres, SQLite, Docker, Git

AI / Agentic: LLM APIs (Anthropic Claude, Gemini), agentic coding & orchestration, prompt engineering, tool-augmented workflows

Data/Systems: High-throughput pipelines, AMPS message queues, concurrency, Pandas, data pipeline design, API integrations, performance optimization

EXPERIENCE

Software Engineer — Bank of America

Aug 2022 – Present

Cirrus OTC Trade Reporting | Python, Pandas, Quartz, AMPS message queues

- Owned production services on the OTC **trade reporting** engine, processing **tens of millions of transactions daily** across the bank's capital markets operations.
- Designed and shipped an extensible **reference-data framework**: orchestrated pipeline of services via message queues querying multiple external APIs per trade event, with retries, load balancing, and error handling.
- Built a **T+1 trade validation service** checking trade-ID uniqueness across counterparties and issuing automated corrections at millions of transactions/day.
- Developed a parallelized **backreporting service**, automating a manual process with significant throughput gains — scoped solution and shipped independently.
- Operated with **high autonomy across teams**, translating requirements from operations and compliance stakeholders into production-grade solutions; optimized service reliability and throughput by orders of magnitude.

Data Analyst (Contract) — Microsoft

Mar 2022 – Aug 2022

Viva Topics | Python, Pandas, Jupyter

- Built data processing and annotation pipelines for **production ML models**; cleaned, transformed, and reported on user-collected datasets for the data science team.

Engineering Intern — Northrop Grumman

Fall 2019, Summer 2020

NX, ANSYS

- Released 50+ engineering drawings including new designs and large assemblies; performed structural analyses for flight and test components.

PROJECTS

- Benchmark Agency Platform** (Next.js, React, TypeScript, Supabase, Python, Anthropic API) — Technical platform for a consulting agency with **paying clients**. Multi-tenant **financial-intelligence dashboard** (revenue, equipment P&L, staff comp, **AI-generated recommendations**) with row-level security. Lead-gen pipeline: discovery, web enrichment, deficiency scoring, and personalized outreach via **Claude**. Embedded with founder — rapid prototyping from real user feedback.
- Market Data Dashboard** (Postgres, Docker, Twilio, Grafana) — Ingests live data from **multiple financial APIs** (Binance, Hyperliquid); automated **alerting pipeline** with SMS notifications and monitoring dashboards.
- Reading Recommendations App** (Flask, Gemini API, SQLite) — **LLM-powered** recommender with prompt engineering, API orchestration, and error handling over a Project Gutenberg corpus.

EDUCATION

Cornell University, College of Engineering — Ithaca, NY

Sep 2017 – Dec 2021

B.S. Computer Science & B.S. Mechanical Engineering | GPA 3.7

- CUAIR (Cornell Unmanned Aircraft)** — Built systems for an autonomous plane with obstacle avoidance, object classification, and airdrop; consistently top team at AUVSI SUAS competition.